William Jin

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Education

University of Waterloo | Bachelor of Software Engineering, Honours *Waterloo, ON | Sep 2024 – May 2028*

- Certifications: Microsoft Azure Fundamentals (AZ-900) and Azure AI Fundamentals (AI-900)
- Relevant Coursework: Data Structures, Software Design (ADTs), Databases, Digital Computers, Statistics

Experience

Software Engineer | Food For Good *Toronto, ON | Sep 2026 – Present*

- Designed and built a volunteer scheduler for a food charity, with role-scoped SQL access and an append-only audit log
- Made **50+** cron jobs and their emails editable by ops, with a simulator that fires any job in a rolled-back transaction
- Fixed SQL injection, account takeover and cross-account authorization bypasses in the legacy production API

AI Engineer | Correctional Service of Canada *Ottawa, ON (Remote) | Jan 2026 – May 2026*

- Built a chatbot on Azure OpenAI and AI Search, letting **17,000+** officers and staff query **220+** policy directives
- Raised search accuracy **15%** and answer grounding **10%** on 300 prompts by re-parsing PDFs to HTML on Azure

Software Developer | SolShare (Solana Foundation Grant) *Oct 2025 – Dec 2025*

- Built Solana's first official native Swift SDK on a **$40K** grant: a typed iOS client for wallet connection and signing
- Secured each wallet session with an X25519 deeplink handshake in TweetNaCl, so the app never sees a wallet key
- Layered a GraphRAG documentation agent over the SDK, indexing **600+** Swift symbols into a FalkorDB code graph

Software Engineer | UAlberta Energy Mechatronics Lab *Edmonton, AB | Jul 2025 – Sep 2025*

- Cut **200+ hours** of manual author lookups by automating citation tracking for 7,000+ papers with a Selenium scraper
- Consolidated **5,000+** fuel cell experiment files into a SQL Server database, replacing manual folder search

Full Stack Engineer | Jindon International Ltd. *Waterloo, ON | Jun 2025 – Jul 2025*

- Built case search and intake forms in Java for a government benefits platform serving **~1M** users
- Optimized Oracle SQL query performance **10x** by eliminating redundant joins and adding targeted indexes

Projects

autobrowse | Personal Project npm *Sep 2026 – Present*

- Built a REST facade over **20** sites that uses each site's own API when it has a token, else drives a Playwright browser
- Turned one-off agent runs into deterministic workflows that self-repair and replay with human-like mouse and typing
- Designed a durable Restate engine where each step chains the next, so pauses and approvals never block a run

Stitch | UofTHacks 13 *Jan 2026*

Winner: Best Use of Gemini API

- Built an agentic video editor in Next.js where a Gemini agent drives a live timeline through **14** custom editing tools
- Implemented natural-language clip retrieval with Twelve Labs, resolving each query to exact in-clip timestamps
- Designed a React 19 timeline where agent and manual edits share a single undo/redo history
- Backed the editor with Supabase Postgres and storage buckets, exporting cuts through FFmpeg in Next.js API routes

Skills

AI/ML: PyTorch, Hugging Face, Unsloth, LangChain, LangSmith, Agno, NumPy, Pandas

Languages: Python, TypeScript, JavaScript, Java, Swift, C++, SQL

Full Stack: React, Next.js, Tailwind CSS, SwiftUI, Node.js, Express.js, Hono, PostgreSQL, MySQL, Supabase

Cloud & Infra: AWS (Lambda, EC2, S3), Cloudflare Workers, Azure, Terraform, Docker, GitHub Actions, Restate

Concepts: Backend Development, Full-Stack Development, Distributed Systems, Data Structures & Algorithms, Object-Oriented Programming, REST APIs, CI/CD, Machine Learning, RAG, AI Agents